explAIn

Retrieval, memory and fine-tuning

Before choosing a mechanism, ask: is this information, state or behavior?

"Information, state and behavior are different problems."

Ask on every project

1. Is this information to know, state to track, or behavior to shape?
2. How often does this knowledge change, and who updates the source?
3. Must the answer be exact and live, or is the closest match good enough?
4. Who may see each document or memory, and is that filtered before retrieval?
5. When an answer is wrong, can we see which chunks were retrieved?

Rules of thumb

- Retrieve facts that change or need a citation; don't bake them into the weights.
- Read orders, balances and permissions from the database of record, never from memory.
- Combine meaning-based and keyword search when users look up codes, names or IDs.
- Fine-tune only after prompting falls short and you have labeled examples and an eval set.

Red flags

- ⚠ Fine-tuning proposed to teach the model the company's documents
- ⚠ A wrong answer blamed on the model before anyone checked what was retrieved
- ⚠ Access permissions applied after retrieval, or not at all
- ⚠ Stored memory that users can't see, correct or delete

Key terms

LoRA	A cheap fine-tune: train small added weights, leave the base model alone.
Chunking	Splitting documents into passages that can be found and read on their own.
RAG	Retrieval-augmented generation: search first, then answer from what was found.
Application memory	State the system stores and re-sends, like preferences or summaries of past chats.
Fine-tuning	Further training on your examples; it changes the weights, and so the behavior.

DECISION GUIDE

Pick the mechanism

If it's a stable rule, tone or short style guide	→ **prompt, with good and bad examples**
If it's documents that change or need sources	→ **retrieval with citations and metadata filters**
If it's per-user preferences or context across sessions	→ **application memory the user can see**
If it must be exact and live (orders, balances, permissions)	→ **database, via code or a tool**
If prompts can't hold a format or narrow task at your volume and cost	→ **fine-tune, with evals**
If it spans several	→ **combine them, but name the primary mechanism and who owns it**